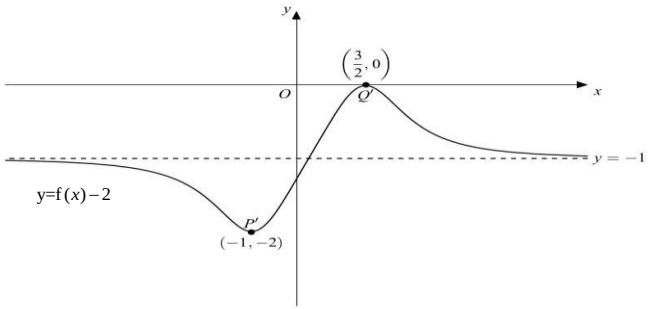
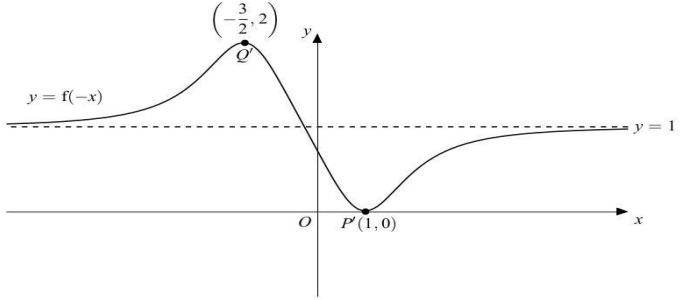
1	$\int 10x^5 + 6x^3 - \frac{3}{x^2}\,\mathrm{d}x = 10 \times \frac{x^6}{6} + 6 \times \frac{x^4}{4} - 3 \times \frac{x^{-1}}{-1}(+c)\,.$	**M1** **A1**
	$= \frac{5x^6}{3} + \frac{3x^4}{2} + \frac{3}{x} + c$	**A1A1**
		(4)

2(a)	$AB = 21\,\text{cm}$, $BC = 13\,\text{cm}$, $\angle BAC = 25°$, $\angle ACB = x°$	
	$\dfrac{\sin x°}{21} = \dfrac{\sin 25°}{13}$ o.e	**M1**
	$\sin x° = 0.6827$ (awrt)	**A1**
		(2)
(b)	$\sin^{-1}(0.6827) = ...$ (43.05°)	**M1**
	($AC < AB$ so $\angle ABC < \angle ACB$ so) required angle is $180° - \sin^{-1}(0.6827) = ...$	**M1**
	So x = awrt 136.95	**A1**
		(3)

3(i)	Writes $\sqrt{180}$ as $6\sqrt{5}$ or $\sqrt{80}$ as $4\sqrt{5}$	**M1**
	Concludes working $\dfrac{6\sqrt{5}-4\sqrt{5}}{\sqrt{5}}=\dfrac{2\sqrt{5}}{\sqrt{5}}=2$	**A1**
		(2)
(ii)	$\dfrac{4\sqrt{5}-5}{7-3\sqrt{5}}=\dfrac{4\sqrt{5}-5}{7-3\sqrt{5}}\times\dfrac{7+3\sqrt{5}}{7+3\sqrt{5}}=...$	**M1**
	$=\dfrac{28\sqrt{5}-35+12\left(\sqrt{5}\right)^2-15\sqrt{5}}{49-9\times5}$	**dM1**
	$=\dfrac{25}{4}+\dfrac{13}{4}\sqrt{5}$	**A1**
		(3)

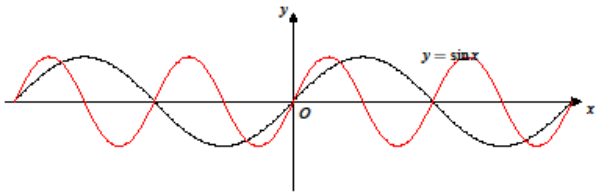
4(i)		Correct shape, translated down.	**B1**
		Correct horizontal asymptote labelled	**B1**
		Correct **maximum** and **minimum** points labelled	**B1**
			(3)
(ii)		Correct shape, reflected in *y* axis	**B1**
		Correct horizontal asymptote labelled	**B1**
		Correct **maximum** and **minimum** points labelled.	**B1**
			(3)

5(a)	E.g. $12-a(x+2)^2$ or $a(x-1)(x+5)$ or $y=ax^2+bx+c \Rightarrow 4a-2b+c=12$ and $25a-5b+c=0$	**M1A1**
	E.g. $0=12-a(-5+2)^2 \Rightarrow a=...$ or $12=a(-2-1)(-2+5) \Rightarrow a=...$ or $2a(-2)+b=0, 25a-5b+c=0, 4a-2b+c=12 \Rightarrow a=..., b=..., c=...$	**dM1**
	$12-\frac{4}{3}(x+2)^2$ or $-\frac{4}{3}(x-1)(x+5)$ or $-\frac{4}{3}x^2-\frac{16}{3}x+\frac{20}{3}$ oe	**A1**
		(4)
(b)	Gradient of l_2 is $\frac{-5}{4}$ o.e.	**M1**
	Equation of l_2 is $y="-\frac{5}{4}"(x+5)$	**M1**
	$y=-\frac{5}{4}x-\frac{25}{4}$	**A1**
		(3)
(c)	For two of $y \geqslant "-\frac{5}{4}x-\frac{25}{4}"$; $y \geqslant \frac{4}{5}x$; $y \leqslant "-\frac{4}{3}x^2-\frac{16}{3}x+\frac{20}{3}"$ Or with strict inequalities	**M1**
	For all three of $y \geqslant "-\frac{5}{4}x-\frac{25}{4}"$; $y \geqslant \frac{4}{5}x$; $y \leqslant "-\frac{4}{3}x^2-\frac{16}{3}x+\frac{20}{3}"$ Or with strict inequalities	**A1ft**
		(2)

6(a)	$2xy-3x^2=50;\ \ y-x^3+6x=0$	
	$\Rightarrow 2x\left(x^3-6x\right)-3x^2=50$	**M1**
	$\Rightarrow 2x^4-12x^2-3x^2-50=0\Rightarrow 2x^4-15x^2-50=0*$ CSO	**A1***
		(2)
(b)	$\Rightarrow \left(2x^2+5\right)\left(x^2-10\right)=0\Rightarrow x^2=...$	**M1**
	So $x^2=10$	**A1**
	$\Rightarrow y=\left(\sqrt{10}\right)^3-6\sqrt{10}=...$	**M1**
	one solution pair is $x=\sqrt{10}, y=4\sqrt{10}$	**A1**
	Solutions are $x=\sqrt{10}, y=4\sqrt{10}$ and $x=-\sqrt{10}, y=-4\sqrt{10}$ CSO	**A1**
		(5)

7(a)	$f'(x) = 2x^{-\frac{1}{2}} + Ax^{-2} + 3 \Rightarrow f''(x) = ...x^{-\frac{1}{2}-1} + -...x^{-2-1}$	**M1**
	$\Rightarrow f''(x) = 2\times -\frac{1}{2}x^{-\frac{3}{2}} + -2Ax^{-3} = -x^{-\frac{3}{2}} - 2Ax^{-3}$	**A1**
	$f''(4) = 0 \Rightarrow -4^{-\frac{3}{2}} - 2A\times 4^{-3} = 0 \Rightarrow A = ...$	**dM1**
	$-\frac{1}{8} - \frac{2A}{64} = 0 \Rightarrow A = -4$	**A1**
		(4)
(b)	$f(x) = \int 2x^{-\frac{1}{2}} + Ax^{-2} + 3\,dx = \frac{2x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + \frac{Ax^{-2+1}}{-2+1} + 3x(+c)$	**M1**
	$= 4x^{\frac{1}{2}} - \frac{A}{x} + 3x(+c)$	**A1ft**
	$f(12) = 8\sqrt{3} \Rightarrow 4\sqrt{12} - \frac{A}{12} + 36 + c = 8\sqrt{3} \Rightarrow c = ...$	**dM1**
	$c = 8\sqrt{3} - 4\sqrt{12} - 36 - \frac{4}{12} = -\frac{109}{3}$ or follow through $c = \frac{A}{12} - 36$	**A1ft**
	So $\left(f(x)\right) = 4x^{\frac{1}{2}} + \frac{4}{x} + 3x - \frac{109}{3}$ oe	**A1**
		(5)

8(a)	With θ being the angle subtended by arc *AB* and ϕ being the angle subtended by arc *CD*	
	$15 = 9 \times \theta \Rightarrow \theta = \frac{5}{3} = (1.67)$	**M1**
	Therefore $\phi = \frac{2\pi}{3} - \frac{5}{3} = (0.4277...)$	**dM1**
	So length of arc *CD* = $84 \times \left(\frac{2\pi}{3} - \frac{5}{3} \right) = 35.929... = 35.9\,\text{cm}$ (1 d.p.)* CSO	**A1***
		(3)
(b)	Perimeter $= 3 \times (15 + 35.9...) + 6 \times (84 - 9)$	**M1**
	= awrt 603 cm (602.787…)	**A1**
		(2)
(c)	FOR EXAMPLE Area of a "blade" is $\frac{1}{2} \times 84^2 \times "\left(\frac{2\pi - 5}{3} \right)"$ = awrt (1510)	**M1**
	Area of sector of inner circle between "blades" is $\frac{1}{2} \times 9^2 \times "\frac{5}{3}"$ = (67.5)	**dM1** **A1**
	Total area is $3\left(\frac{1}{2} \times 84^2 \times "\left(\frac{2\pi}{3} - \frac{5}{3} \right)" + \frac{1}{2} \times 9^2 \times "\frac{5}{3}" \right) = ...(4729.577764\,\text{cm}^2)$	**ddM1**
	So area is awrt 0.473 m^2 or awrt 4730 cm^2	**A1**
		(5)

9(a)	(i) $2p$		**B1**
	(ii) $-p$		**B1**
	(iii) $3-p$		**B1**
			(3)
(b)		Correct shape, same height starting at *O*, scaling may be incorrect. Two repeats of the sin*x* graph each side	**M1** **A1**
			(2)
(c)	For $x=\frac{\alpha}{2}$		**B1**
	Attempt at second root E.g. $x=\frac{180°-\alpha}{2}$		**M1**
	$x=90°-\frac{\alpha}{2}$		**A1**
			(3)

10(a)	$\frac{dy}{dx} = \frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}$	**M1 A1**
		(2)
(b)	At $x = -\frac{7}{2}$, $\frac{dy}{dx} = \frac{6}{7}\left(-\frac{7}{2}\right)^2 + \frac{2}{7}\left(-\frac{7}{2}\right) - \frac{5}{2} = ...(=7)$	**M1**
	So at B we know $\frac{dy}{dx} = "-\frac{1}{7}"$	**M1**
	hence $\frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2} = -\frac{1}{7}$	**dM1**
	$\Rightarrow 12x^2 + 4x - 35 = -2 \Rightarrow 12x^2 + 4x - 33 = 0$*	**A1***
		(4)
(c)	E.g. $12x^2 + 4x - 33 = 0 \Rightarrow (2x-3)(6x+11) = 0 \Rightarrow x = ...$	**M1**
	From graph we can see the x coordinate is positive, so $x = \frac{3}{2}$ at B	**A1**
		(2)
(d)	Equation of l is $y = "-\frac{1}{7}"x - 1$	**M1**
	Finds coordinates of A $x = -\frac{7}{2} \Rightarrow y = "-\frac{1}{7}" \times -\frac{7}{2} - 1 = \left(-\frac{1}{2}\right)$	**dM1**
	Substitutes $x = -\frac{7}{2}, y = "-\frac{1}{2}"$ into $y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k \Rightarrow k = ...$	**ddM1**
	$k = \frac{5}{4}$ CSO	**A1**
		(4)

Alt I (d)	At A $y=\frac{2}{7}\left(-\frac{7}{2}\right)^3+\frac{1}{7}\left(-\frac{7}{2}\right)^2-\frac{5}{2}\left(-\frac{7}{2}\right)+k=...\left(=k-\frac{7}{4}\right)$	**M1**
	Equation of l is $y-\left(k-\frac{7}{4}\right)=-\frac{1}{7}\left(x+\frac{7}{2}\right)$ or y intercept is $"-\frac{1}{7}"\times"\frac{7}{2}"+"k-\frac{7}{4}"$	**dM1**
	$\Rightarrow y=-\frac{1}{7}x+k-\frac{9}{4}\Rightarrow k-\frac{9}{4}=-1\Rightarrow k=...$	**ddM1**
	$k=\frac{5}{4}$	**A1**